

SPOKANE FELTS FIELD, WA, USA

WMO: 727856

Lat: 47.683N

Lon: 117.321W

Elev: 595

StdP: 94.37

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94176

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-12.5	-9.5	-17.8	0.8	-10.3	-14.5	1.1	-7.9	10.8	5.4	9.1	4.5	1.2	60	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	34.9	17.8	33.0	17.3	31.2	16.8	19.3	31.6	18.3	30.3	17.4	28.8	2.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.0	11.5	21.9	13.8	10.6	21.6	12.7	9.9	20.8	57.0	31.4	53.7	30.2	50.9	28.6	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.7	6.6	-15.9	38.0	3.7	1.6	-18.6	39.1	-20.8	40.1	-22.9	41.0	-25.6	42.2		
(5)			-16.3	21.5	3.7	2.4	-18.9	23.2	-21.1	24.6	-23.1	25.9	-25.8	27.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-0.1	1.3	5.2	8.8	13.7	17.2	22.0	21.3	16.3	9.4	3.6	-0.5		
	DBStd	8.68	4.47	3.97	3.42	3.34	3.85	3.59	3.20	3.10	3.66	3.70	4.19	4.27		
	HDD10.0	1358	314	245	152	62	10	1	0	0	2	55	193	324		
	HDD18.3	3357	572	478	408	287	151	62	8	9	81	279	441	582		
	CDD10.0	1324	0	0	2	26	125	217	373	351	192	35	3	0		
	CDD18.3	282	0	0	0	0	7	29	122	103	21	1	0	0		
	CDH23.3	4169	0	0	0	13	164	462	1701	1437	381	12	0	0		
	CDH26.7	1873	0	0	0	1	42	164	846	681	138	2	0	0		
(14) Wind	WSAvg	2.2	1.7	1.9	2.6	2.7	2.6	2.6	2.5	2.2	2.0	1.9	1.9	1.7		
(15) Precipitation	PrecAvg	606	71	45	71	45	47	49	14	17	27	53	75	83		
	PrecMax	720	166	62	119	77	77	106	31	54	63	174	148	129		
	PrecMin	510	36	15	37	26	27	16	2	3	10	14	46	37		
	PrecStd	63	32	11	24	13	17	24	10	14	13	35	23	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	12.4	18.5	25.2	30.5	34.5	37.5	37.1	33.7	24.8	15.9	12.2		
		MCWB	7.2	7.6	10.0	13.8	16.0	18.5	19.4	18.9	16.8	14.2	12.3	8.6		
	2%	DB	8.4	10.1	15.7	21.6	27.6	30.9	35.2	34.6	30.6	21.0	13.1	8.7		
		MCWB	5.7	6.4	8.9	12.0	15.3	17.0	17.9	17.6	16.1	12.4	9.4	6.3		
	5%	DB	6.9	8.3	13.4	18.4	24.9	28.0	33.5	32.7	27.6	17.9	11.3	6.9		
		MCWB	4.7	5.1	7.7	10.3	13.7	15.9	17.5	17.1	15.4	11.1	8.2	4.8		
	10%	DB	5.3	6.9	11.3	16.0	22.2	26.0	31.7	30.9	25.0	15.9	9.6	5.0		
		MCWB	3.7	4.0	6.7	8.9	12.7	14.8	17.1	16.6	14.1	10.0	7.0	3.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	8.5	11.2	14.7	17.6	19.5	21.1	20.6	18.5	15.2	12.6	8.8		
		MCDB	9.5	11.3	16.2	23.4	27.3	31.7	34.8	33.7	29.2	22.8	15.0	11.7		
	2%	WB	6.1	6.9	9.7	12.5	15.7	17.9	19.6	19.0	17.0	13.3	10.1	6.7		
		MCDB	7.9	9.5	14.4	20.3	25.2	28.7	31.8	31.8	28.0	18.9	12.7	8.7		
	5%	WB	4.9	5.5	8.4	10.9	14.6	16.7	18.6	17.9	15.8	11.9	8.7	5.0		
		MCDB	6.6	7.9	12.5	17.5	23.1	26.6	30.8	30.1	25.8	16.8	10.7	6.6		
	10%	WB	3.8	4.3	7.2	9.6	13.4	15.5	17.6	16.9	14.7	10.6	7.3	3.5		
		MCDB	5.3	6.5	11.0	15.0	20.7	24.2	29.3	28.6	23.6	14.7	9.1	4.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	7.6	10.0	11.5	12.9	13.0	16.1	16.2	14.7	11.2	7.2	5.4		
		MCDBR	7.1	9.7	13.7	16.5	17.6	17.7	19.5	20.0	19.5	15.1	8.9	6.9		
	5% WB	MCWBR	5.1	6.4	7.9	8.2	7.2	6.5	6.3	7.0	7.6	7.6	5.9	5.2		
		MCWBR	6.7	8.4	12.1	15.2	15.7	16.4	18.1	18.3	17.7	13.2	7.6	6.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.277	0.287	0.307	0.316	0.330	0.333	0.333	0.346	0.321	0.307	0.287	0.275		
		taud	2.483	2.512	2.518	2.489	2.473	2.497	2.495	2.446	2.506	2.540	2.546	2.483		
	Ebn at Noon	Ebn at Noon	821	892	922	940	933	929	924	896	893	854	804	776		
		Edh at Noon	63	75	88	100	105	104	102	103	88	73	59	56		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.03	1.96	3.00	4.50	5.80	6.23	7.36	6.11	4.34	2.57	1.22	0.81		
	RadStd	RadStd	0.13	0.27	0.36	0.43	0.47	0.55	0.41	0.41	0.42	0.29	0.13	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.69	N/A	N/A	N/A	+113	+44
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+92	N/A

Nomenclature: See separate page